ANATOMICAL TERMS

Anatomical Positions

1. Erect anatomical position:

- This is standard position used to **describe the anatomy** of the human body.
  - Human body is standing erect.
  - Eyes & face are looking forwards.
  - Upper limbs are hanging by the sides of the body.
  - Palms are facing forwards.
  - Thumbs are directed laterally.
  - The lower limbs are close together to the middle line.

Erect anatomical position

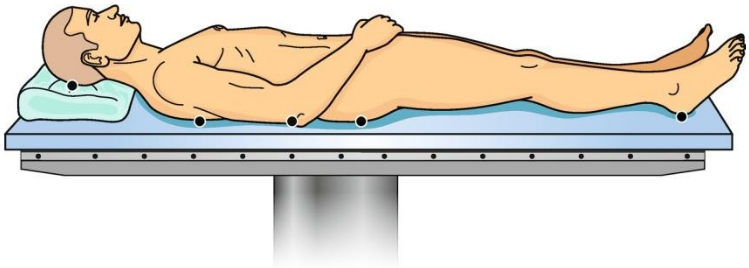
2. Supine position: The body lies on its back.

3. Prone position: The body lies on its face.

4. Lateral decubitus position: The body lies on its side (right or left).

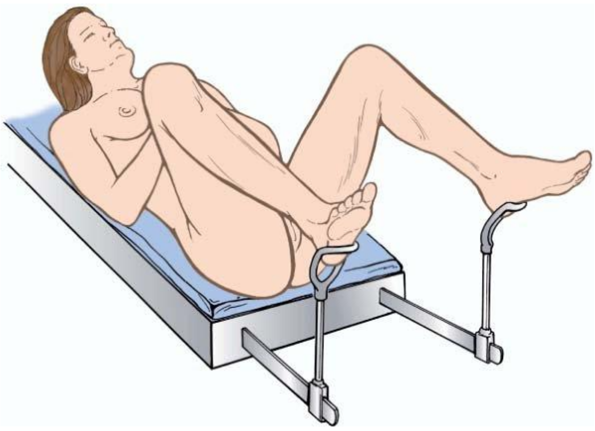
5. Lithotomy position: The body lies on its back with flexion of hip and knee and abduction of hip joints.

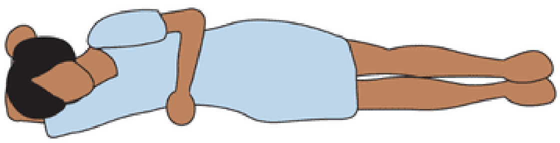
Supine position



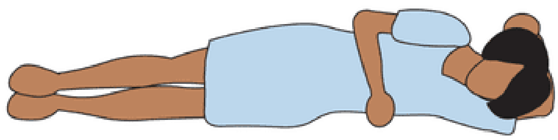
Prone position

Lithotomy position





Right Lateral Recumbent



Left Lateral Recumbent

Anatomical Planes

1. Median (Sagittal) plane:

- It is the vertical plane which passes in the middle line of the body dividing it longitudinally into **equal right and left halves**.

2. Paramedian plane:

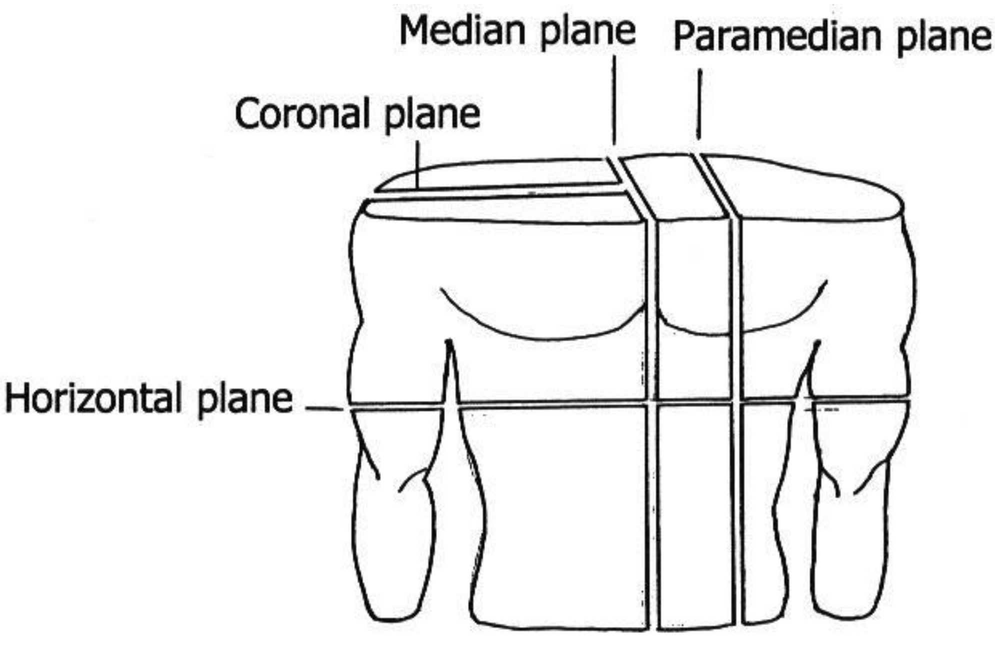
- It is a vertical plane parallel to and **nearby the median plane**.
- It may be **right or left**.
- It divides the body into **2 vertical unequal parts**.

3. Coronal (Frontal) plane:

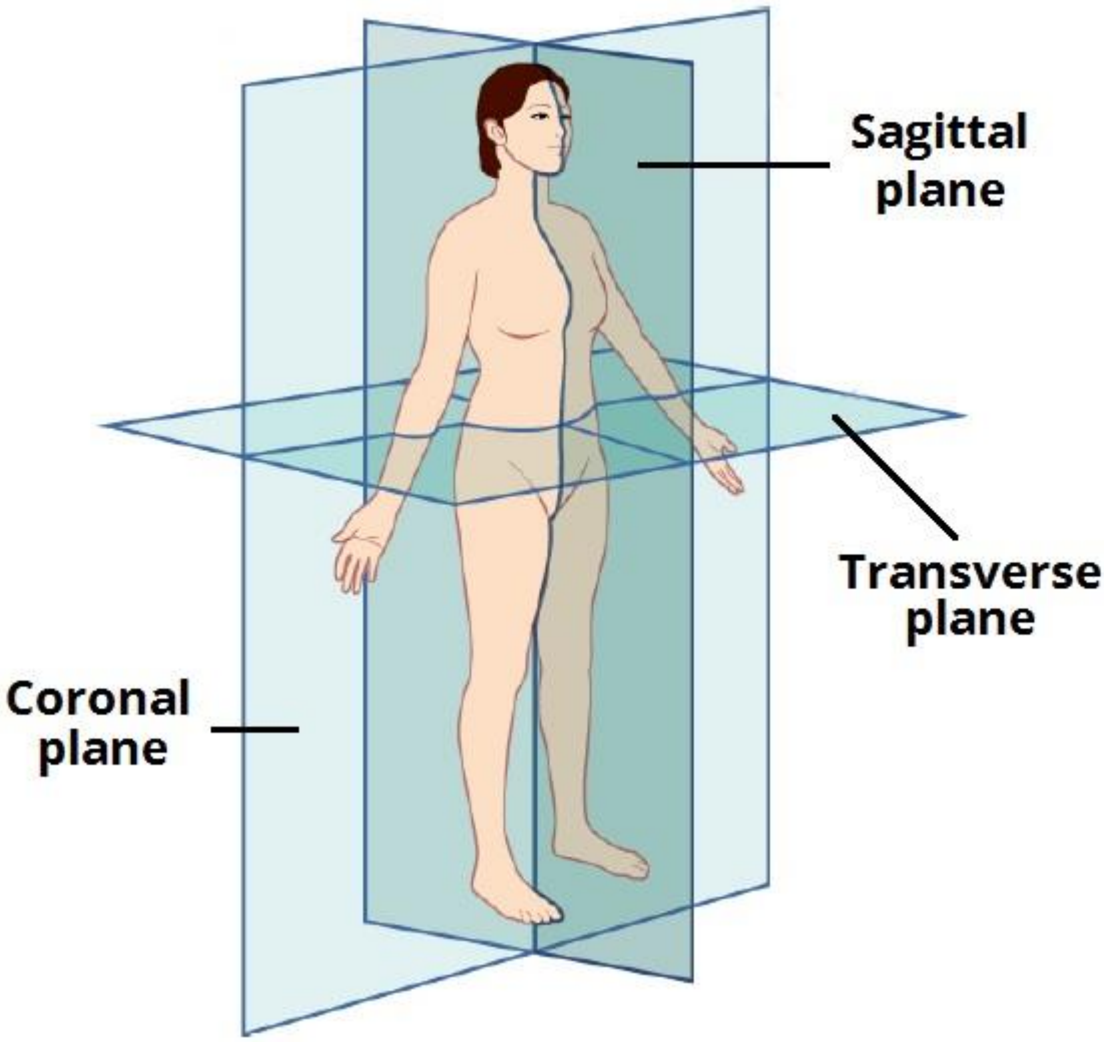
- It is a **vertical plane** which divides the body vertically into an **anterior part** towards the front of the body and a **posterior part** towards the back.

4. Horizontal plane:

- It is the transverse plane which runs horizontally dividing the body into an **upper and lower parts at any level** of the body.



Planes of the Body

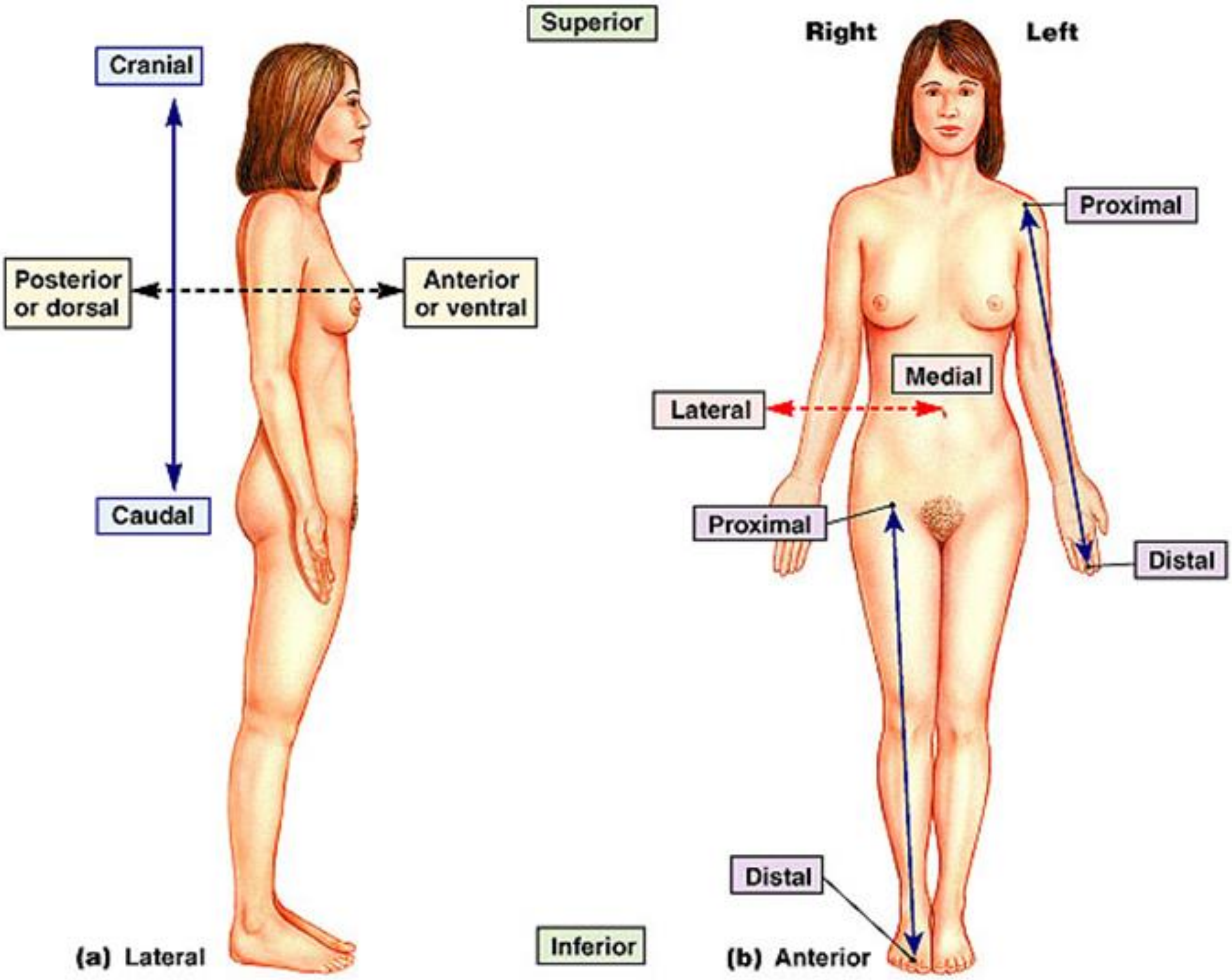


Anatomical Planes

Terms of Position

Term	Meaning
Anterior	In **front** or nearer to the front of the body.
Posterior	**Behind** or nearer to the back of the body.
Superior	Near to the **upper** end of body.
Inferior	Near to the **lower** end of body.
Median	Exactly in the **middle line**
Medial	**Nearer** to the median plane.
Lateral	**Away** from the median plane.
Proximal	Nearer to the **root** of the limb.
Distal	Away from the root of the limb.
Superficial	Towards the skin or **body surface**.
External = Outer	Nearer or on the surface of the body.
Deep	Away from the skin or body surface.

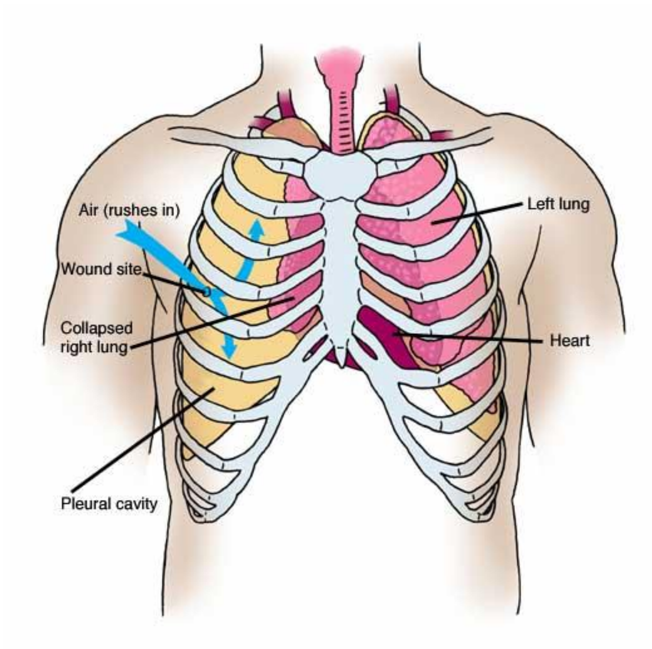
Terms of Position

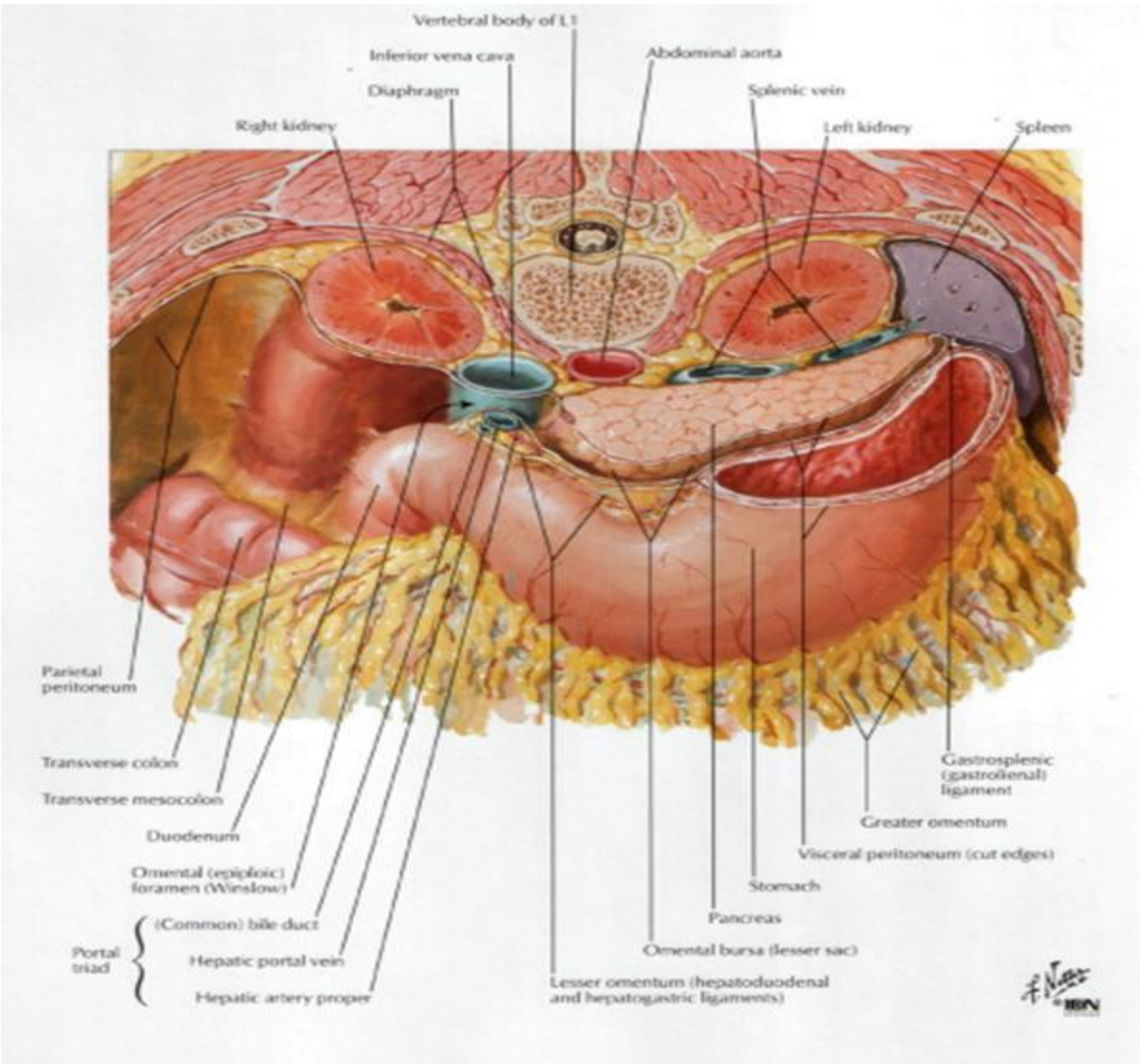


TERMS OF NUMBER

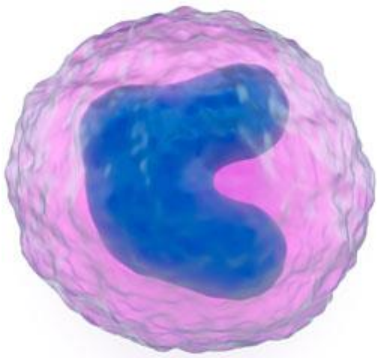
Uni- = Mono-	**One**
Bi- = Di-	**Two**
Tri-	**Three**
Quadri-	**Four**
Multi- = Poly-	**Many**
Oligo-	**Little**

Example for Terms of Position

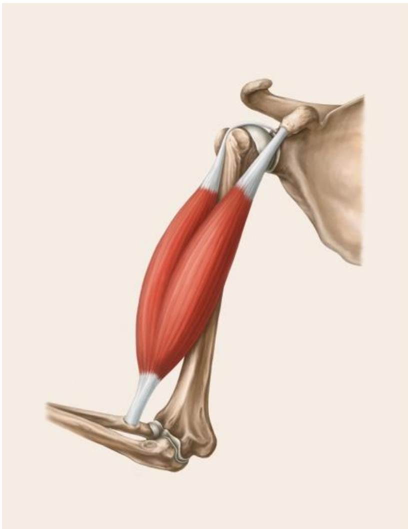




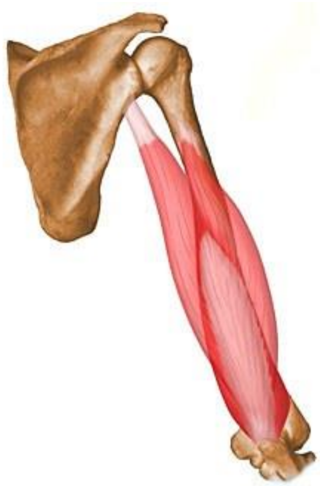
Example for Terms of Number



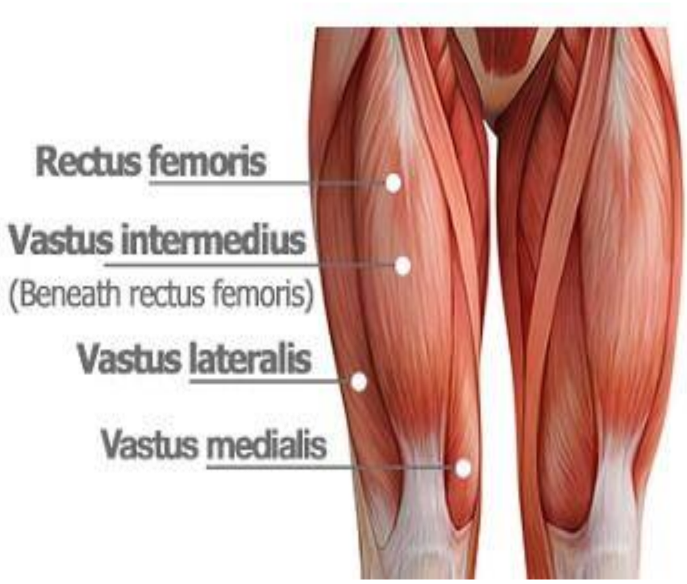
Monocytes



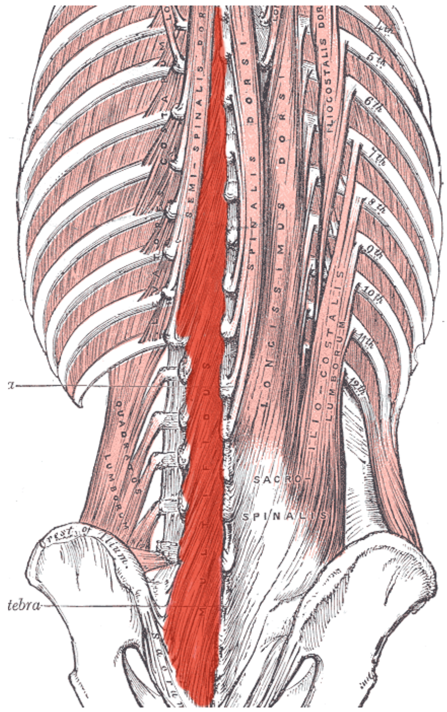
Biceps



Triceps



Quadriceps

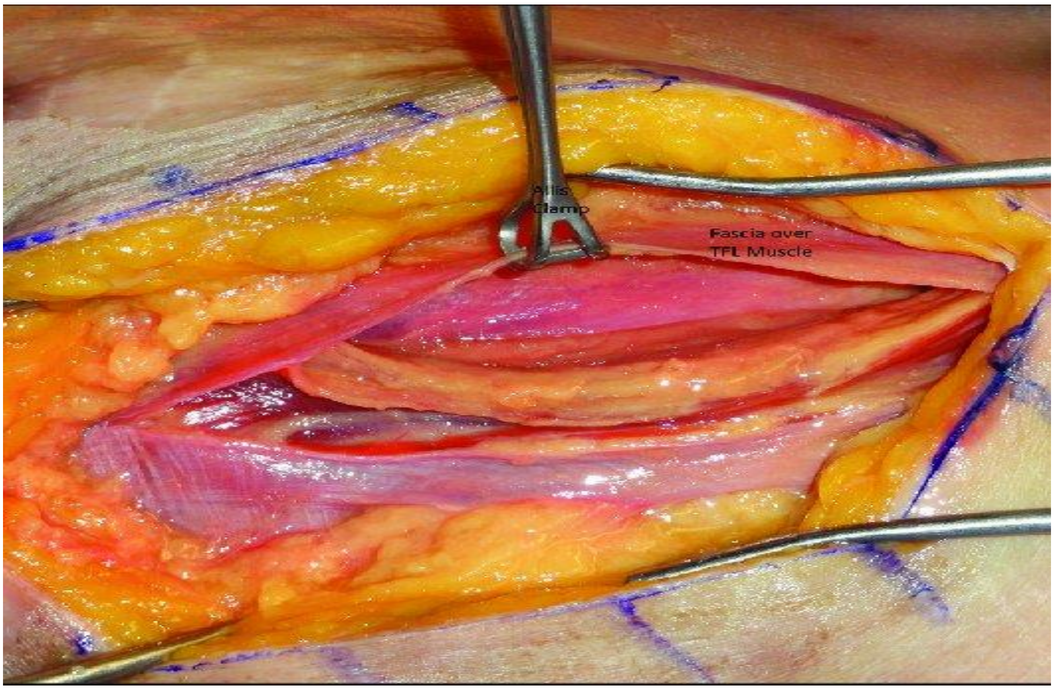


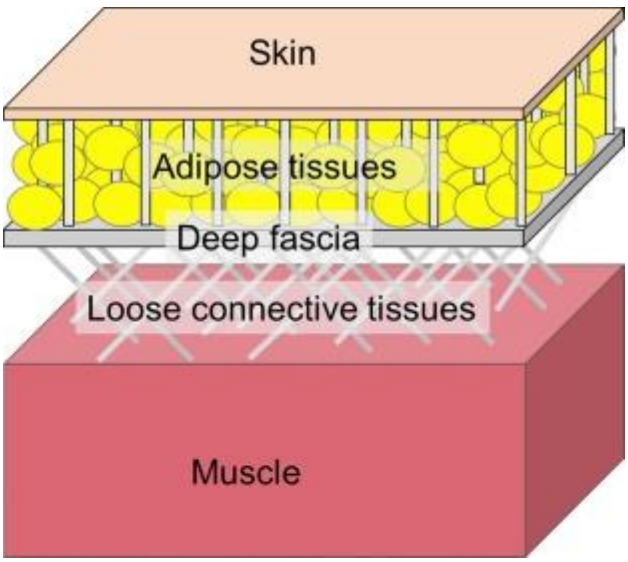
Multifidus

FASCIA

⁎ It is a type of **connective tissue** which have the following features:

- It lies **deep to the skin**.
- It **covers** the body wall and limbs.
- It binds and **connects** different body structures together.
- It is divided into superficial, deep and internal fasciae.





A- Superficial Fascia:

- A layer of **loose connective tissue** lies immediately deep to the skin.
- It connects the skin to the underlying structures.
- The superficial fascia **contains** a variable quantity of **fat** which is more in females than in males.
- **Functions of the superficial fascia:**
  1. It prevents **heat** loss from the body, so it acts as **thermal insulator** and allows **storage of energy** (due to the presence of fat).
  2. It acts as a medium **conducting** the cutaneous **nerves, blood vessels and lymphatics** which supply the skin.

B- Deep Fascia:

- A membrane composed of compact collagen fibers.
- It is inelastic.
- It is well defined in the limbs.
- It is absent in the face and in the anterior wall of abdomen.

CARTILAGE

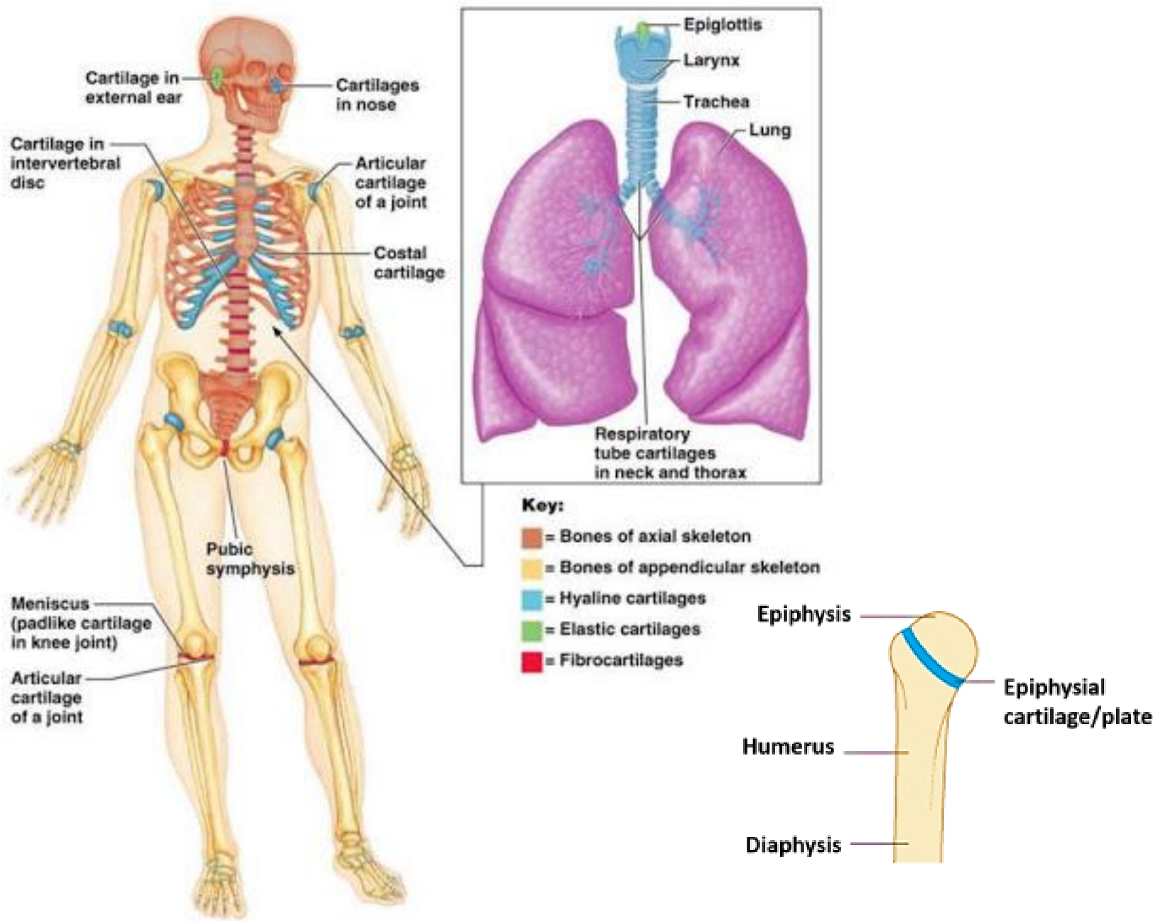
✳ **Properties:**

1. It is a rubbery type of **connective tissue**, it is tough and resilient.
2. It has **no** blood vessels, nerves or lymphatics.
3. Gets its **nutrition** by diffusion from the blood vessels of perichondrium.
4. **It consists of** mature cartilage cells (chondrocytes), fibers and matrix.
5. **Resists** compression forces and friction.

✳ Types of Cartilage:

I) Hyaline Cartilage: (Glass-like)

- **Characters:** Commonest type of cartilage in human body.
- **Matrix:** Translucent.
- **Ossification** by age: Occurs in certain sites e.g. in epiphyseal cartilages.
- **Sites:**
  1) In the developing bones in the **fetus**.
  2) In the **epiphyseal plates** of the long bones.
  3) The **articular** cartilage in joints.
  4) The **costal** cartilage and **xiphoid** process.
  5) **Nasal cartilage**s.
  6) The **larynx** (except the epiglottis) and **tracheal rings**.

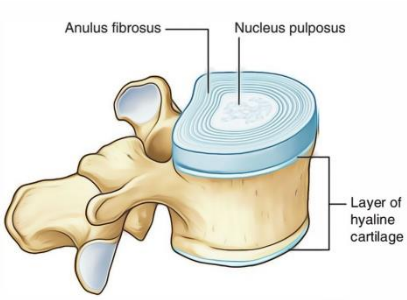


II) White fibrocartilage:

- **Characters:**

  a) **Matrix:** Opaque because it is rich in collagen bundles.

- **Sites:** Intervertebral discs.
- **Ossification in old age:** Does not occur.

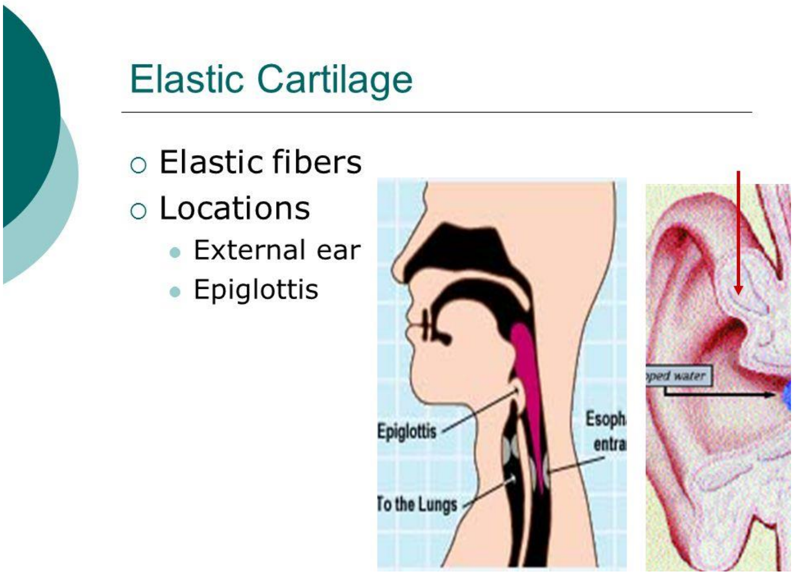


III) Yellow elastic fibrocartilage:

- **Characters:**

  a) **Matrix:** Yellow in appearance because it is rich in yellow elastic fibers.

- **Sites:**
  1. Auricle of the ear.
  2. Epiglottis of the larynx.
- **Ossification in old age**: Does not occur.



BONES

✳ **Definition & properties:**

▪ It is a special type of **hard connective tissue** which forms the skeleton.

✳ **Functions of Skeleton:**

1-Gives the specific **shape** to the body.

2-Provides the **central axis** of the body and the **skeleton** of both upper and lower **limbs**.

3-**Protects** the vital organs: the skull protects the brain, and thoracic cage protects the heart and lungs.

4-Provides surface area for **muscular attachment**.

5- Acts as levers for **muscle contraction & movements**.

6-Transmits and supports the **body weight** e.g. vertebral column transmits the weight of the head and trunk to the bony pelvis then through the bones of lower limbs to the feet and lastly to the ground.

7- Forms the **joints** to make an important part of the locomotor system.

8- Forms the **blood elements** in the red bone marrow.

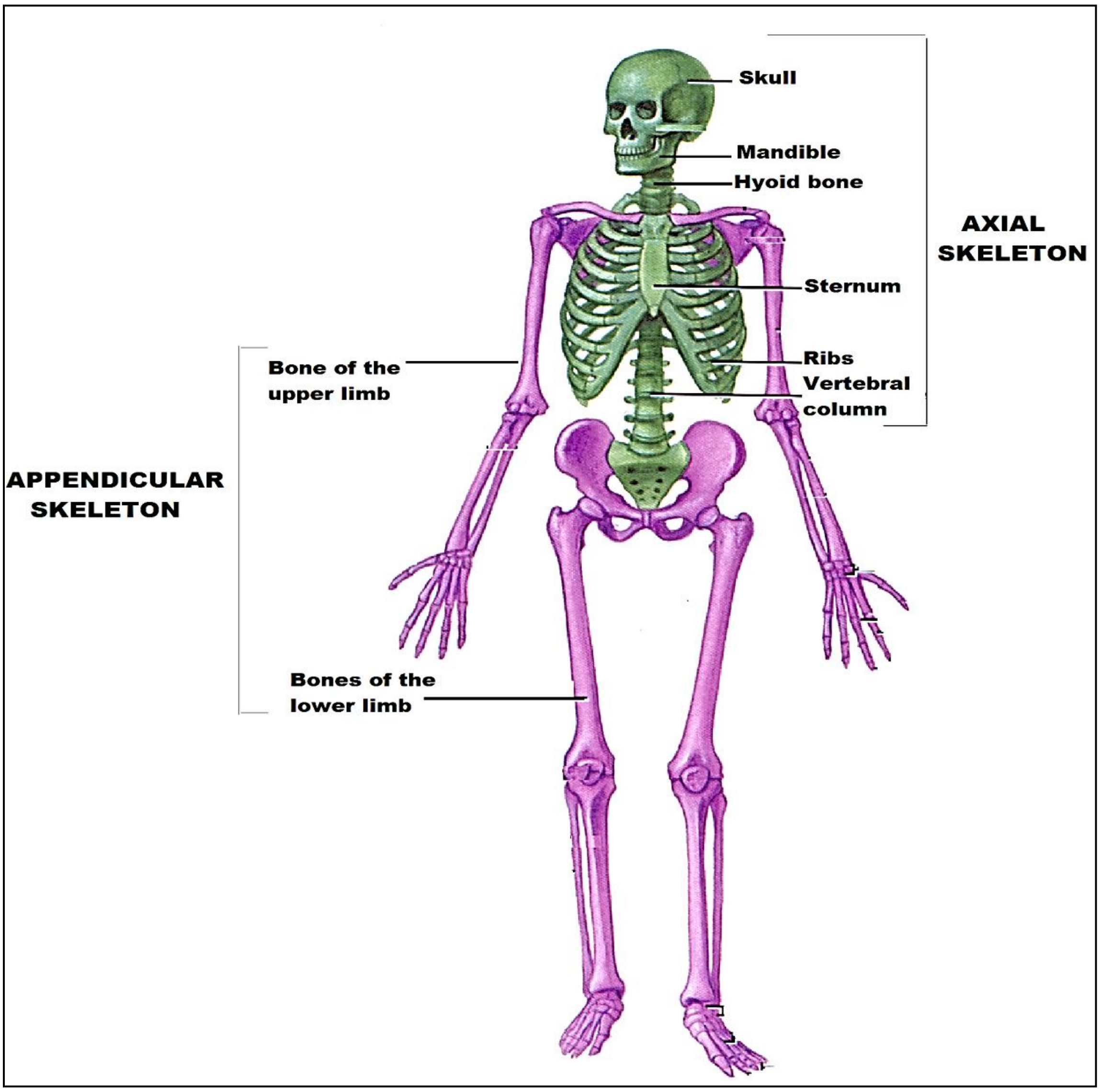
9-Stores **calcium & phosphorus salts**.

✳ **Classification of Skeleton:**

I) According to the position in the body:

1- **Axial skeleton:** Skull, mandible, hyoid, sternum, ribs and vertebral column (i.e bones of head , neck & trunk).

2- **Appendicular (peripheral) skeleton**: in upper and lower **limbs**



III) According to structure of bones :

1- Compact (hard or ivory)bones :

- It is the outer hard mass covering the surface of all bones .

2- Cancellous (spongy) bones :

- It is a **network of trabeculae** lying inside compact bones.
- In between these trabeculae there are many **spaces** filled with bone **marrow** .

IV) According to the shape of bones:

1- Long bones

- **Site:** Present in upper and lower **limbs.**

- **Structure: two ends** (epiphyses) and **a shaft** (diaphysis) in between.

- **N.B :** ***Physis means growth plate .***

a) Epiphysis:

♣ It is the **expanded** upper and lower **ends** of the long bone.

♣ It is used for **articulation** and its articular surface is **covered** with a layer of articular hyaline **cartilage**.

b) Diaphysis (Shaft):

♣ A **tube of compact bone (cortex)** with a **central medullary cavity lined with endosteum** and is **filled with bone marrow** (soft vascular tissue). This meullary cavity **doesn't extend** to the epiphysis or the metaphysis.

♣ The shaft is **covered** with sheath called **periosteum which formed of :**

- **Superficial** layer formed of fibrous tissue.
- **Deep** layer formed of fibrous tissues , osteoblast (bone forming cells) , sensory nerve fibers and blood vessels (supply the underlying bone).

♣ The long bones **increase in diameter** from periosteum.

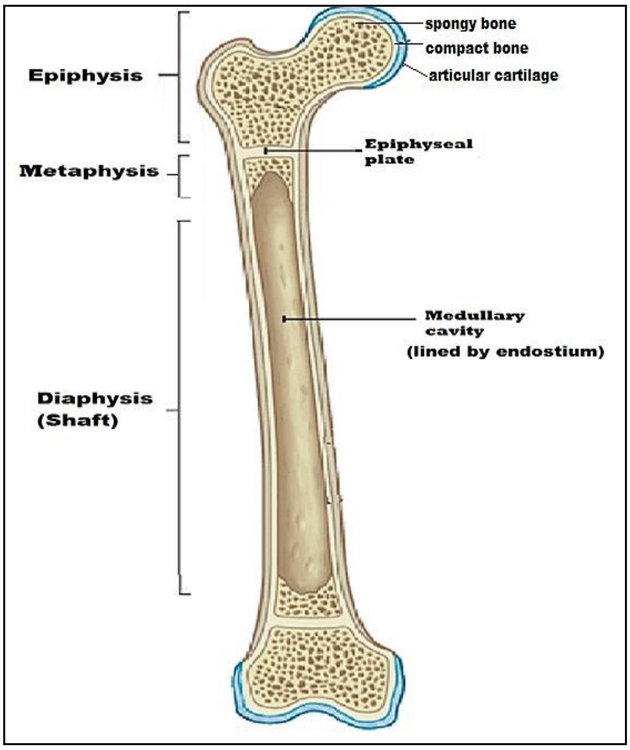
♣ In the **growing** long bones, the epiphysis and the diaphysis are separated by a disc of hyaline cartilage called **epiphyseal cartilage** which is responsible for the **growth in length**.

c) Metaphysis:

♣ It lies in the upper and lower parts of the long bone just **close to the epiphyseal cartilage**.

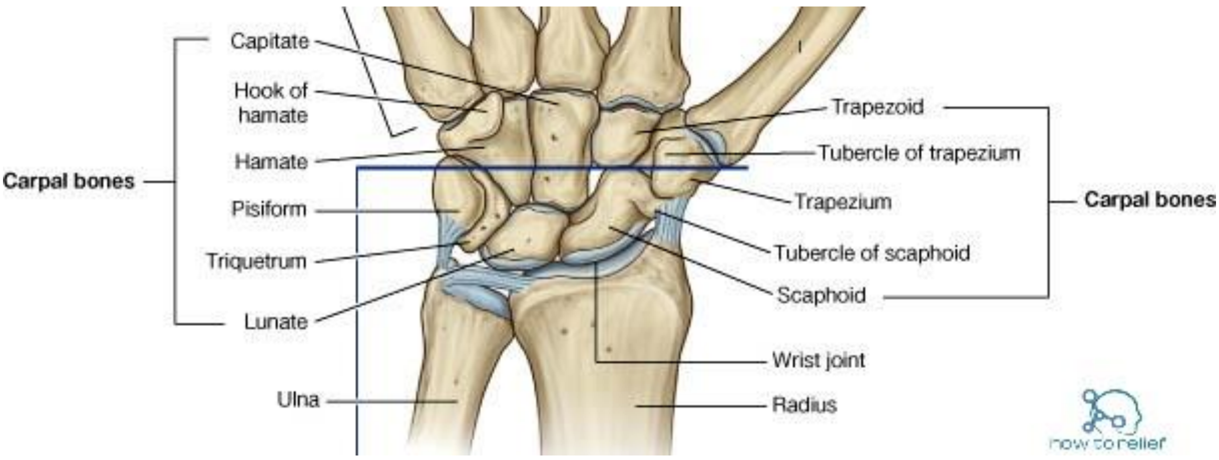
- It is the most **active** part of long bone and contains the **newly formed bone**, formed by the epiphyseal cartilage, which gradually migrate towards the diaphysis.

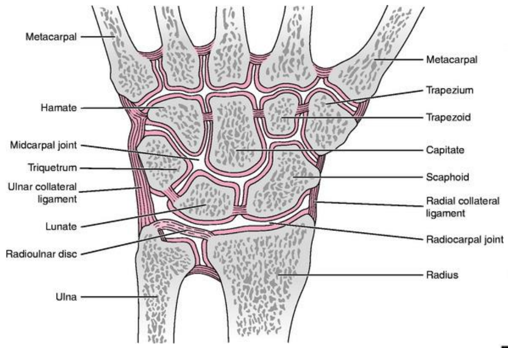
Structure of long bones

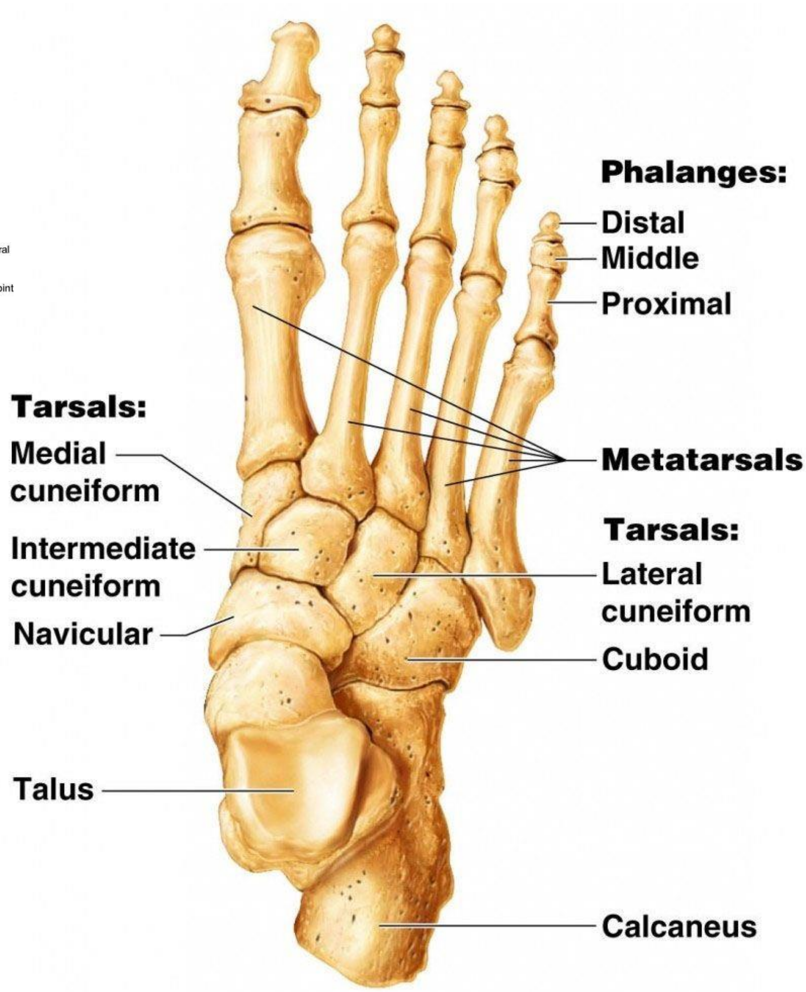


2) Short Bones:

- **Site:** e.g. carpal bones (in hand) and tarsal bones (in foot).
- **Structure:** consist of **spongy** bone covered with thin layer of **compact** bone

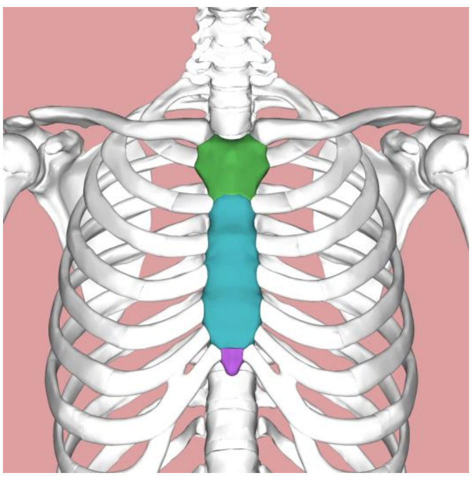


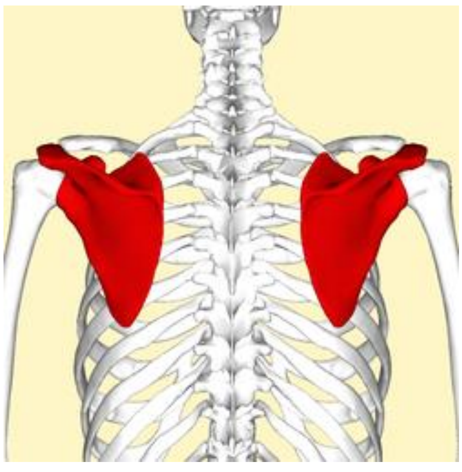


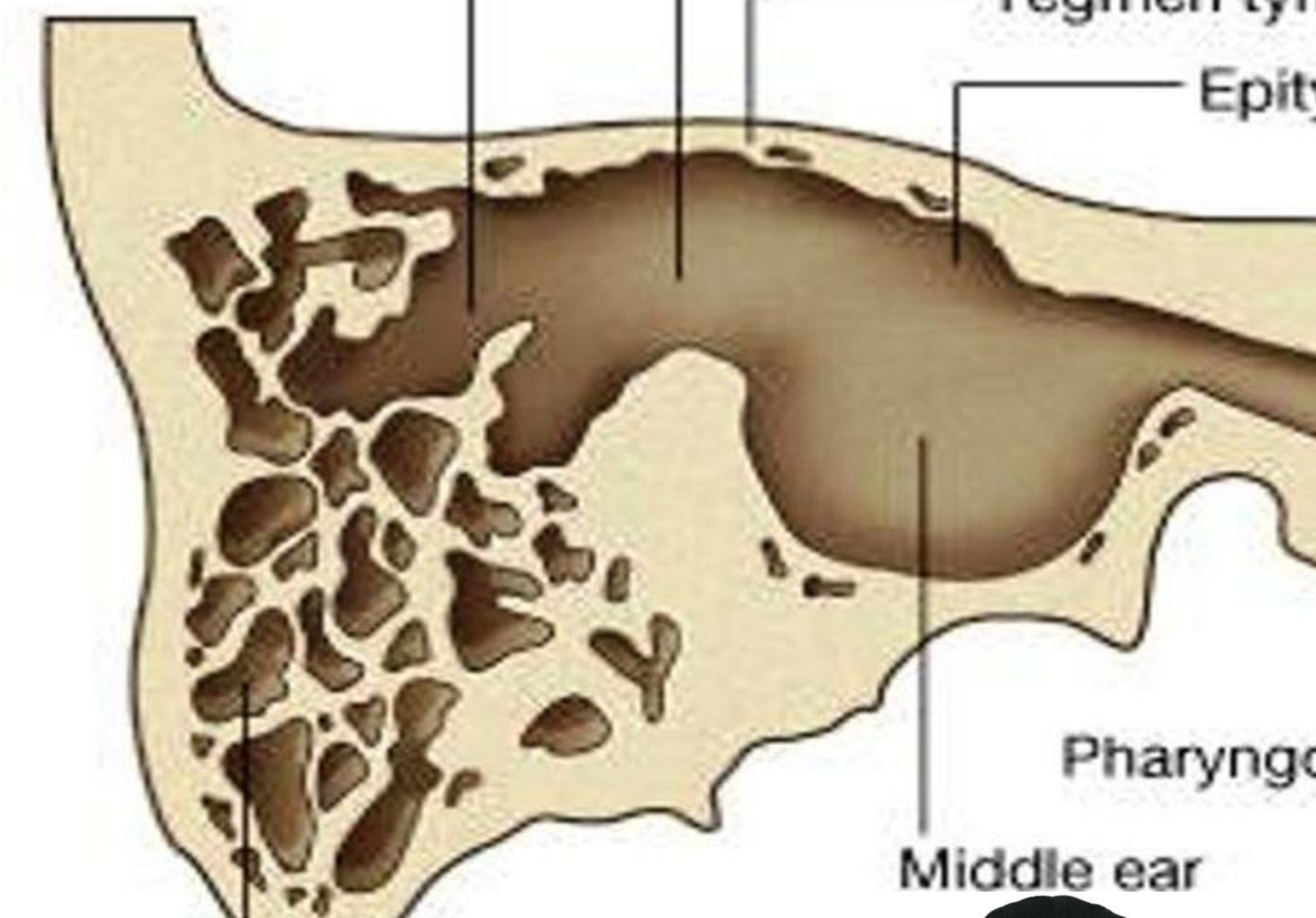


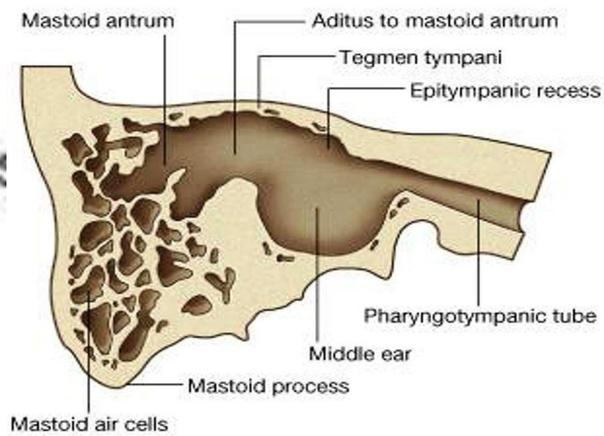
3) Flat Bones:

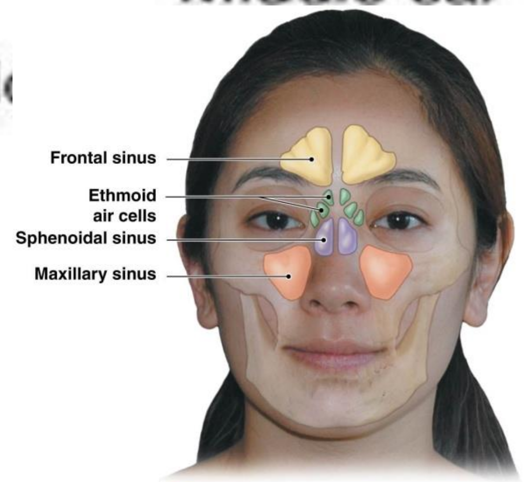
- **Site:** e.g. bones forming **skull cap**, **scapula, sternum and ribs.**
- **Structure:** consist of two thin **plates of compact** bone with middle layer of **spongy** bone.
- **Function:** 1- Protection. 2- Muscular attachment









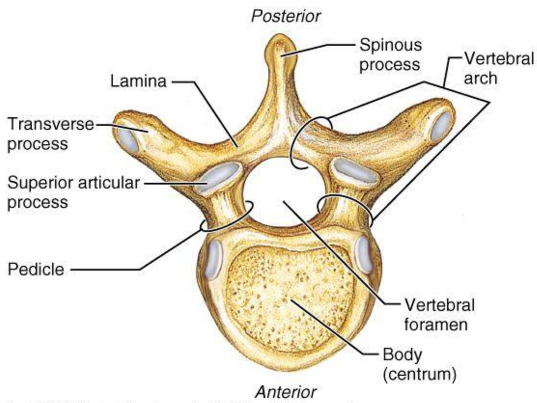


a Locations of the paranasal sinuses.

5- Irregular Bones:

- **Shape and Site:** Bones of **irregular** shape with projecting **processes** e.g. vertebrae.
- **Structure:** They are similar in structure to the short bones.

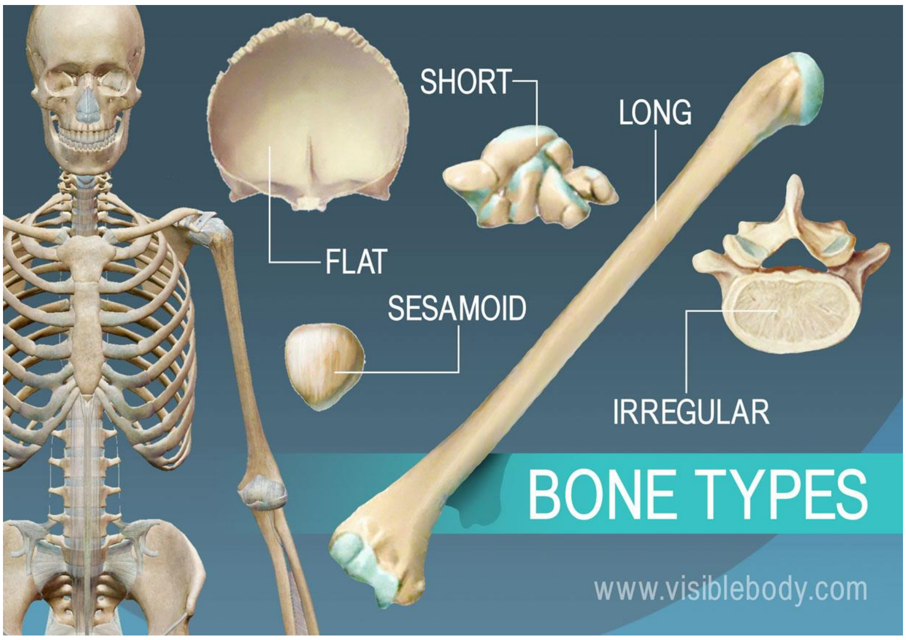
Vertebra



6- Sesamoid Bones:

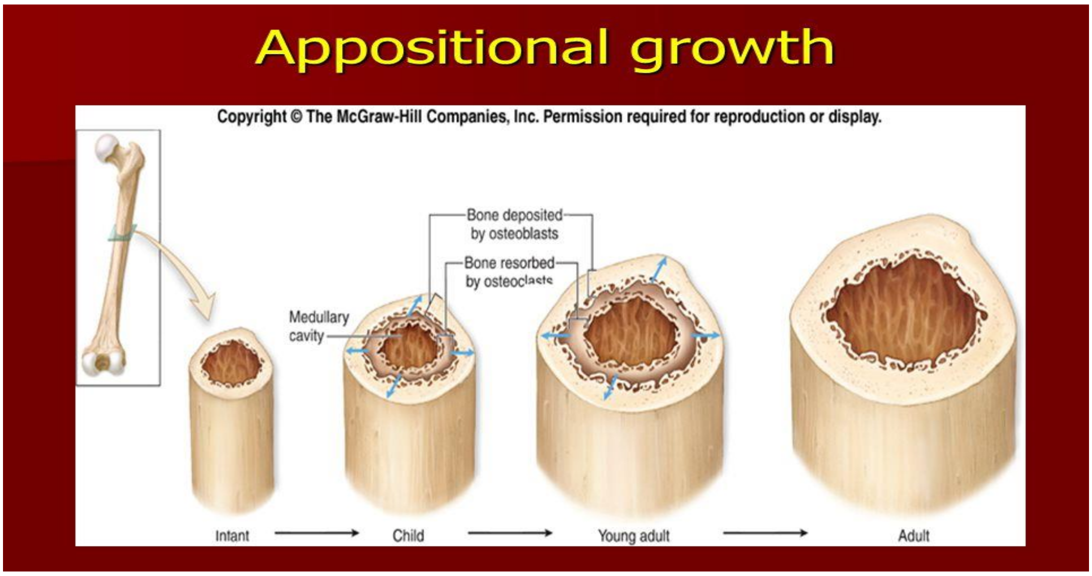
- **Structure:** small **nodules** of bones.
- **Site: embedded** in some muscle tendons e.g. patella (largest sesamoid bone, embedded in the tendon of quadriceps femoris in front of knee joint).
- **Function:** They **diminish friction** between tendons and underlying bones.

> ❖ **N.B: All bones formed of 2 layers of compact bone with a layer of spongy bone in between except diaphysis of long bones and pneumatic bones.**

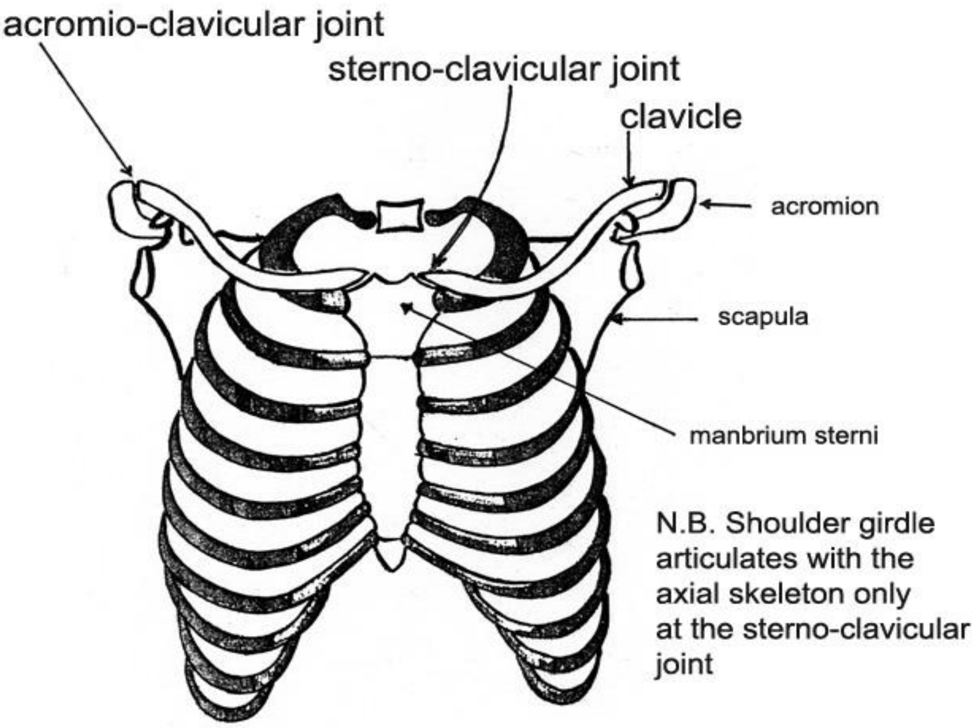


✳ **Growth of bones:**

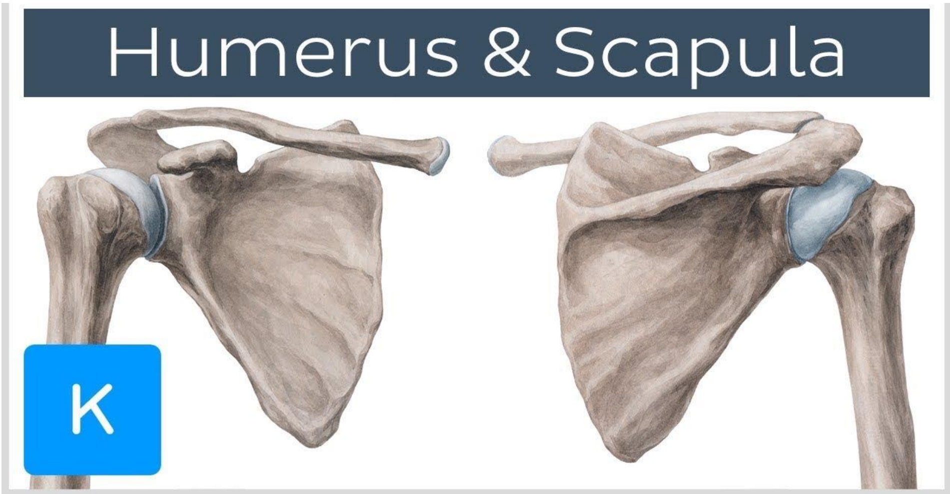
- The long bones **increase in length** from the **epiphyseal cartilages** by proliferation of its cells.
    - ♣ When the bone becomes **mature**, the cells of the epiphysial plate stops division and ossifies resulting in **fusion** between epiphysis and diaphysis.
    - ♣ Fusion of the epiphysis and diaphysis is under **hormonal** control.
    - ♣ It occurs in **females earlier** than males by about 2 years.
- The long bones **increase in width** from **osteoblasts in the periosteum around** the external bone surface. At the same time, osteoclasts **in the** endosteum break down bone on the internal bone surface, around the medullary cavity.

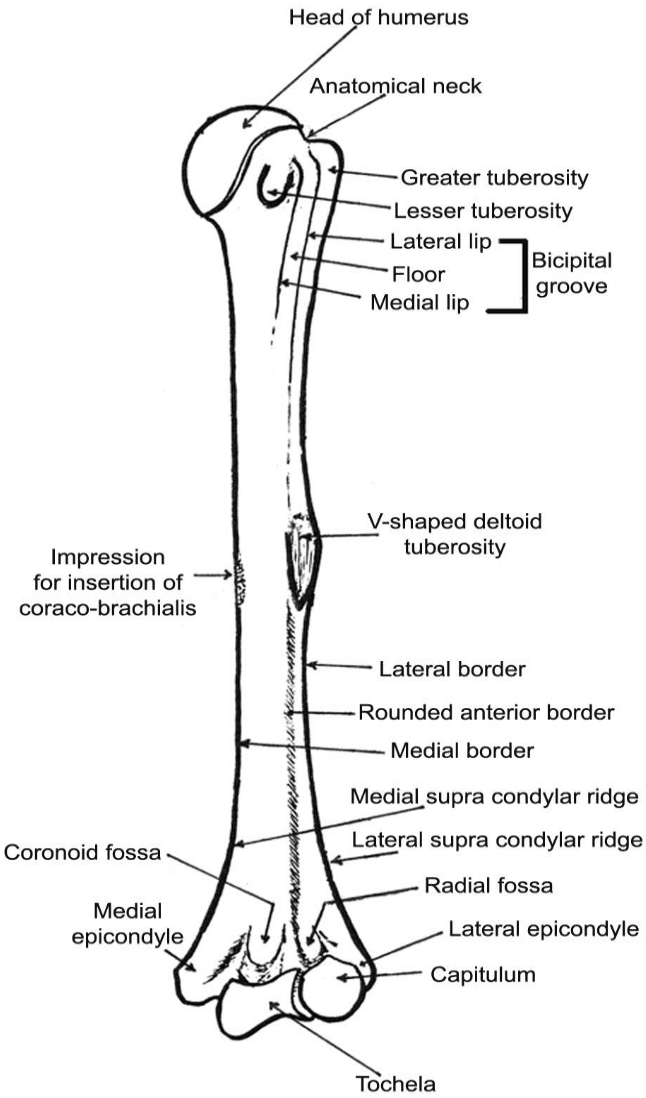


I) Bone of Upper Limb



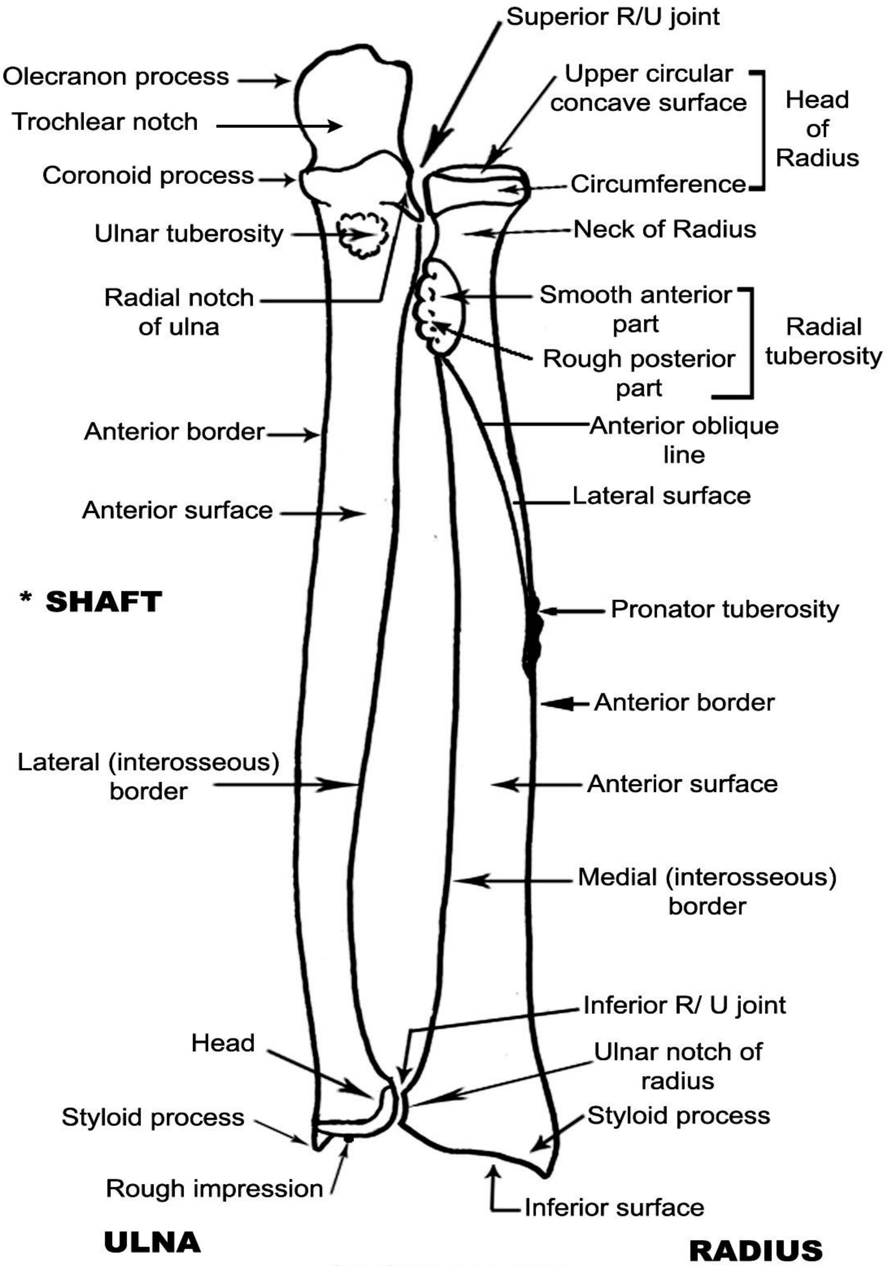
Shoulder Girdle



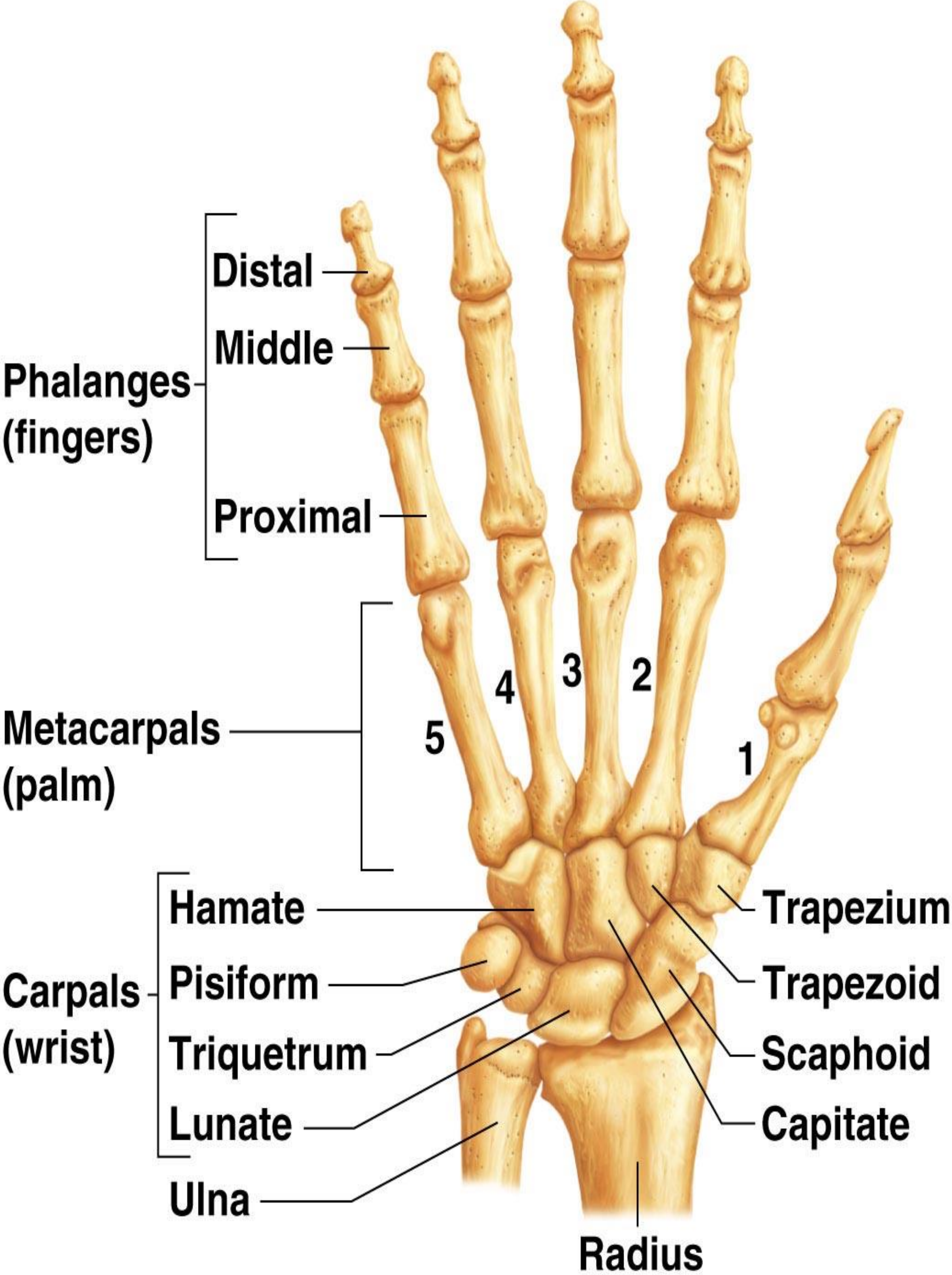


HUMERUS
(Anterior view)

* UPPER END

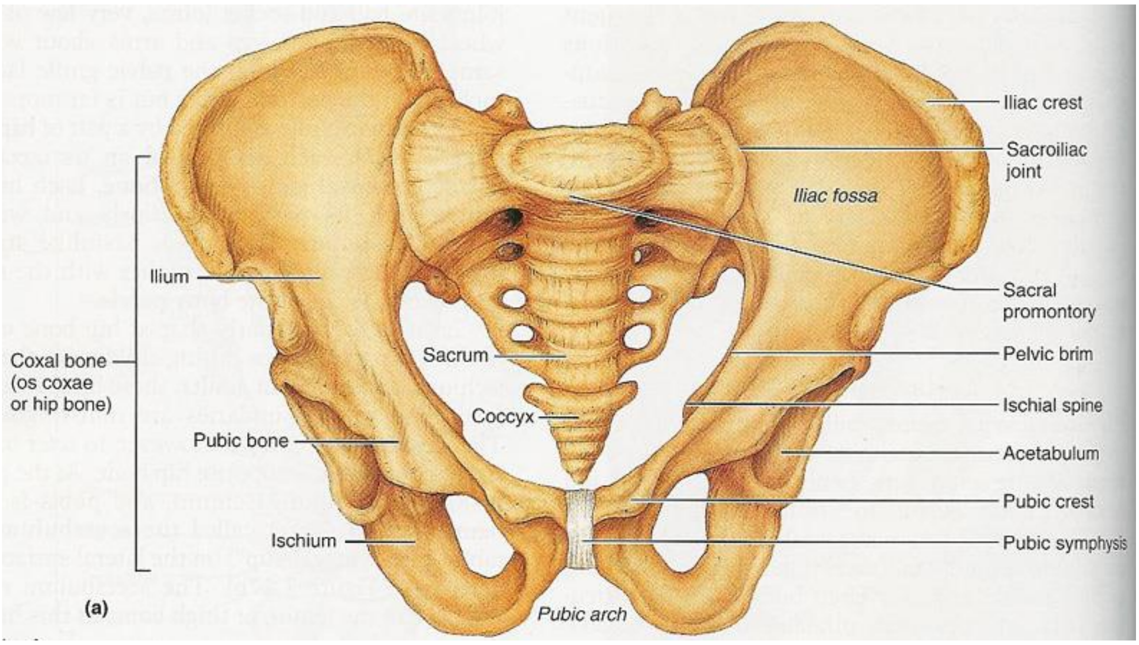


* LOWER END
(Anterior view)

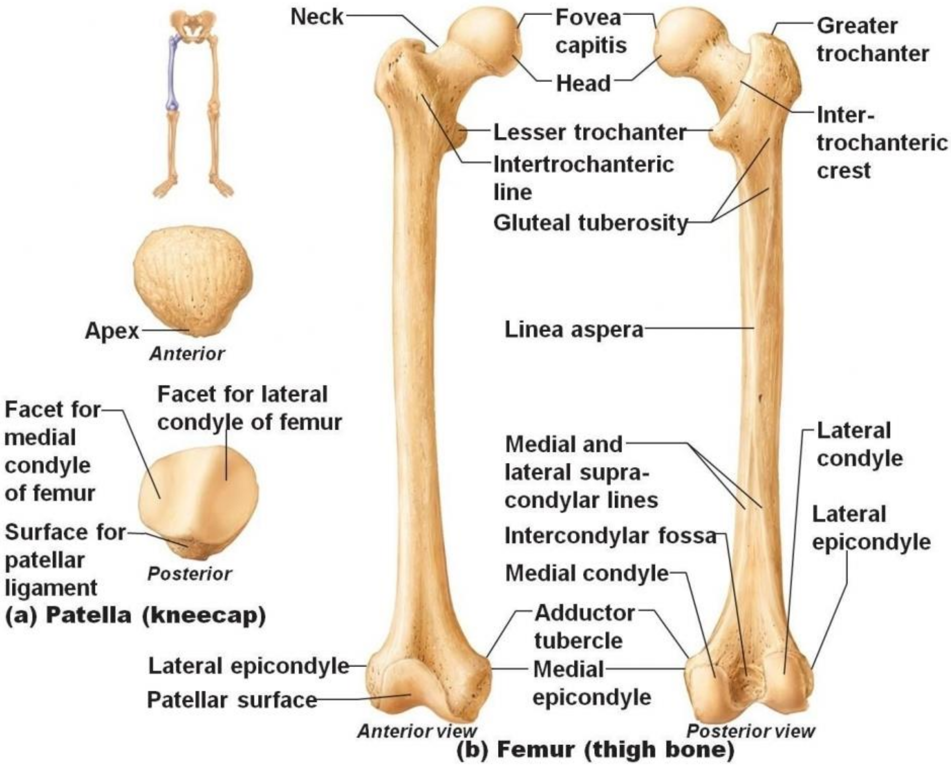

Phalanges (fingers)
Distal
Middle
Proximal
Metacarpals (palm)
5
4
3
2
1
Carpals (wrist)
Hamate
Pisiform
Triquetrum
Lunate
Ulna
Trapezium
Trapezoid
Scaphoid
Capitate
Radius

Bones of lower limb

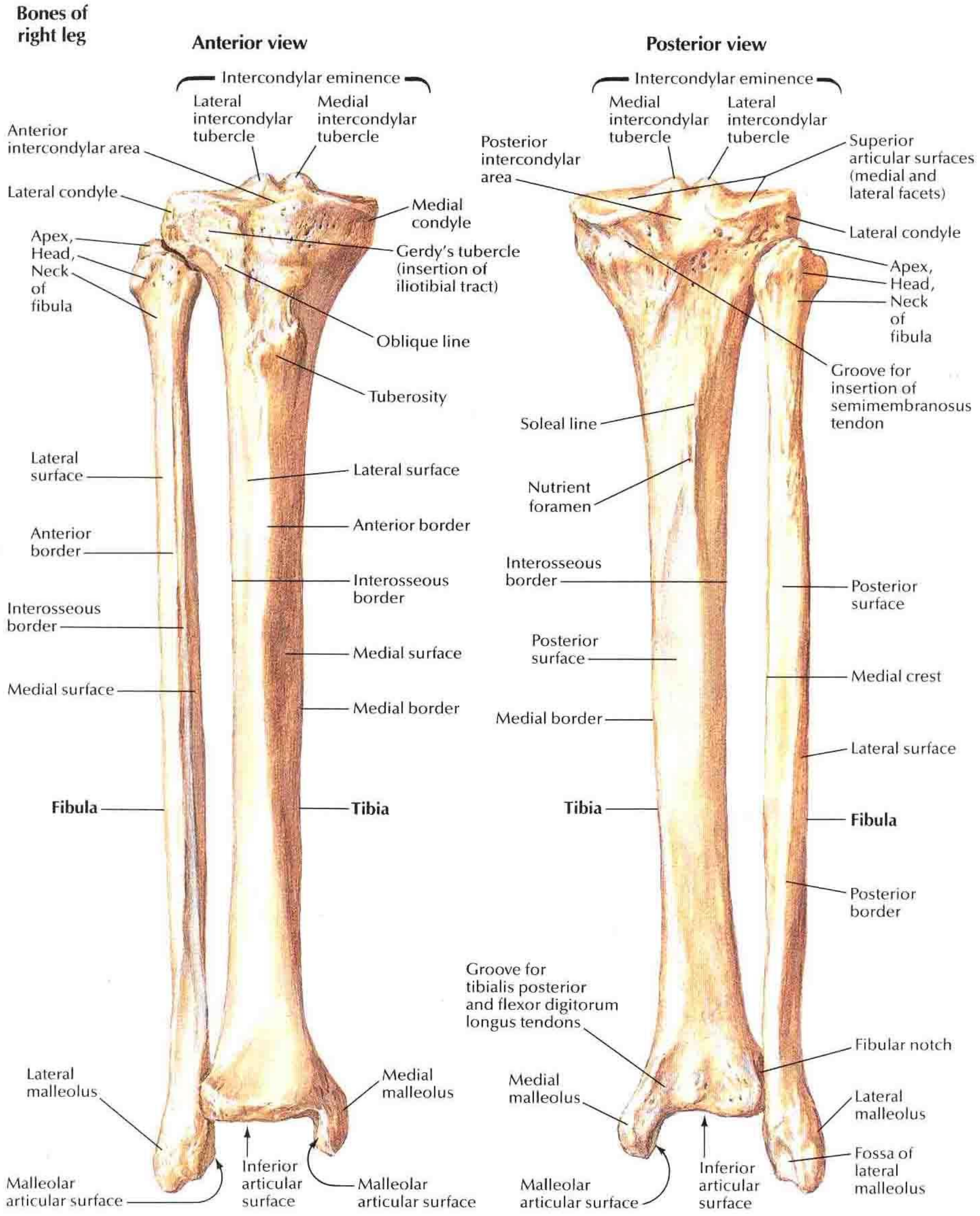
Hip Bone



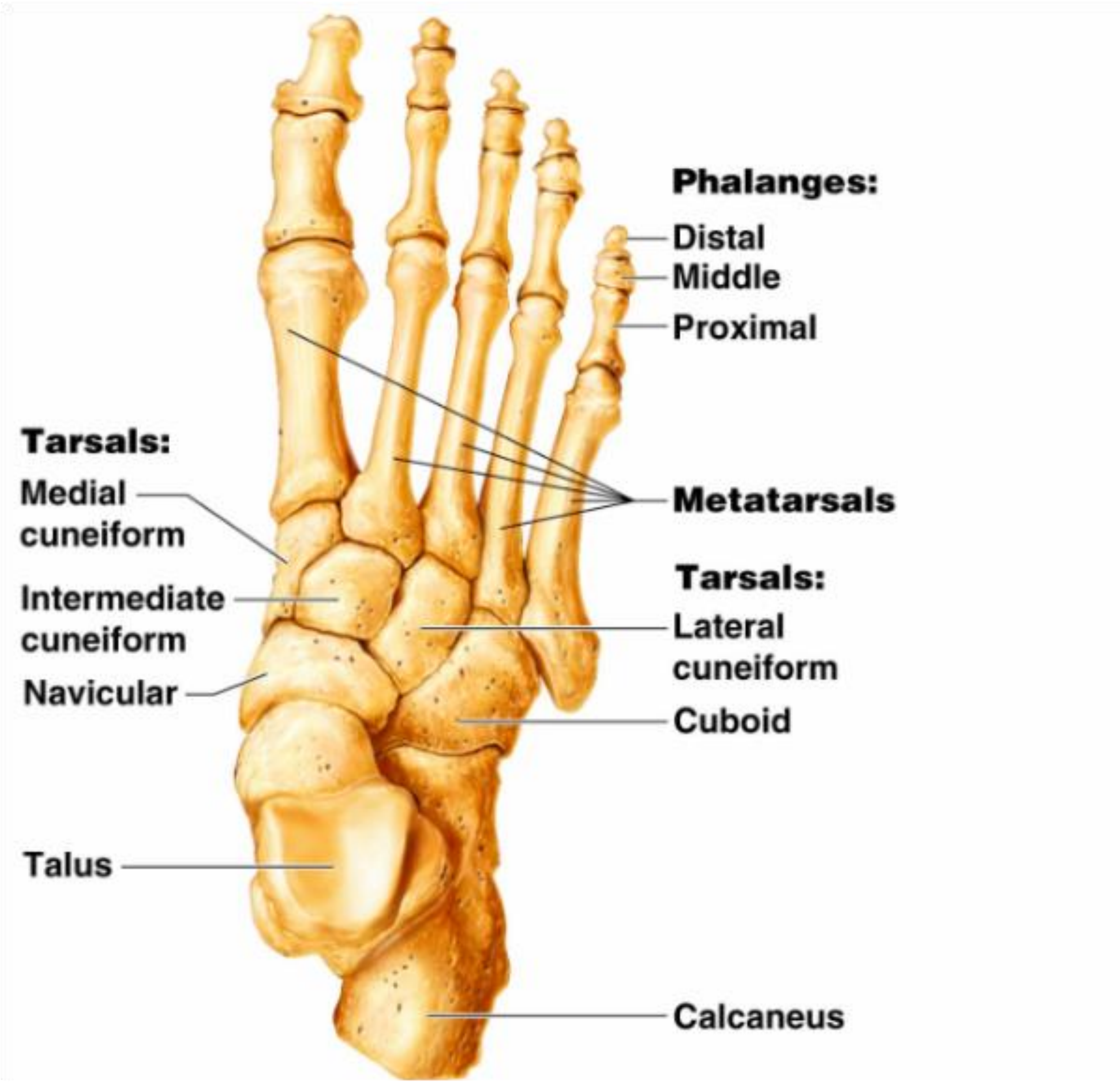
Femur and Patella



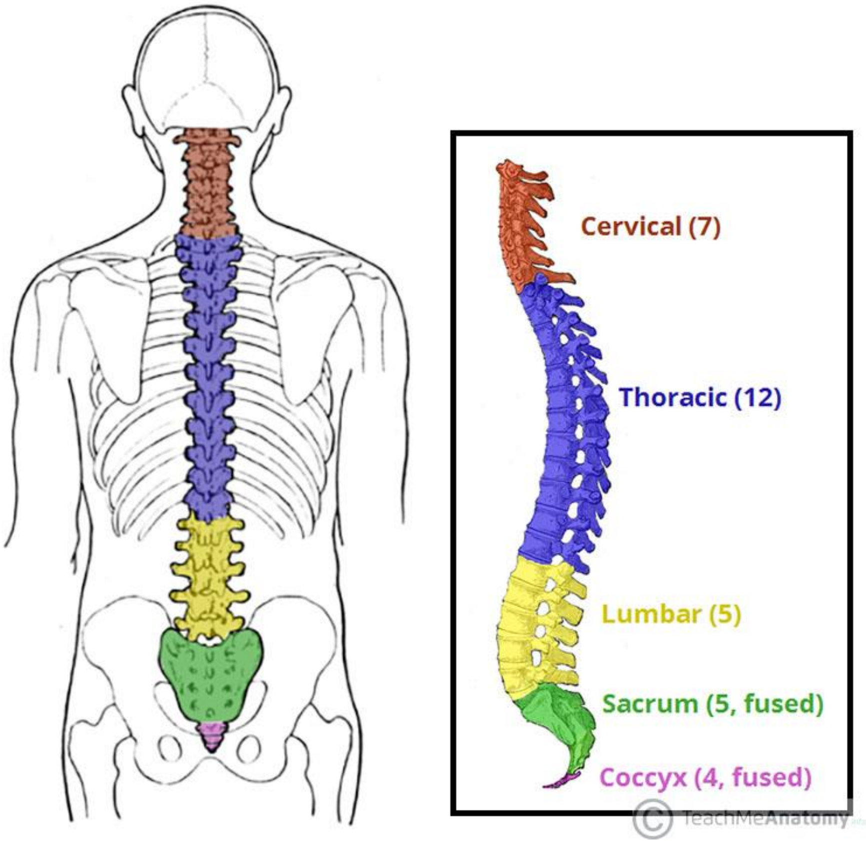
Tibia and Fibula

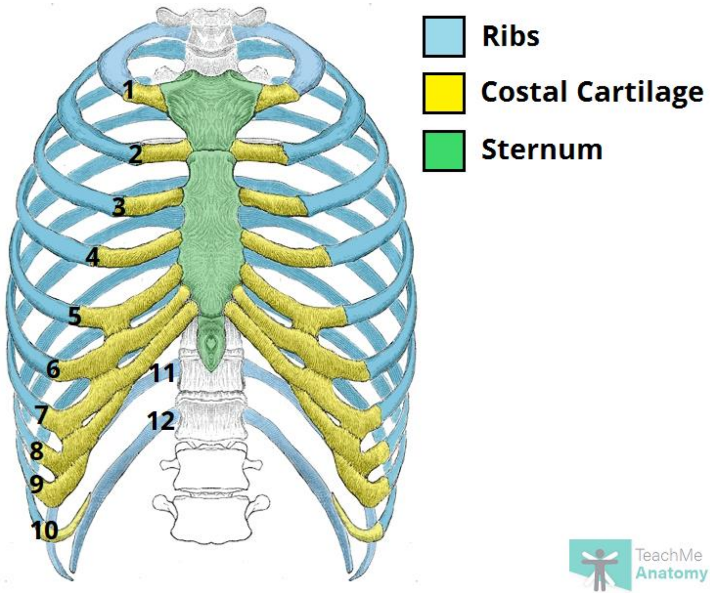


Bones of the foot



Vertebral column





Ribs and Sternum